Mühendislik Fakültesi
Bilgisayar Mühendisliği

2024-2025 Güz Dönemi

Yapay Sinir Ağları

Proje Ödevi

Zehirli Yılan Sınıflandırma

	1. Öğrenci	2. Öğrenci	3. Öğrenci
Okul No	**214210093**	**214210040**	**214210095**
Ad”	**Oğuzhan**	**Muhammet Akif**	**Erkan**
Soyad	**YALÇIN**	**AYAN**	**YİĞİT**
Ders Hocası	**Ömer Faruk AKMEŞE**		

2024
Çorum / Türkiye

z39dhpm2t

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#GEREKLİ KÜTÜPHANELERİN YÜKLENMESİ

[12]:
```
import numpy as np
import os
import pandas as pd
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow.keras.models import Model
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Dropout
from tensorflow.keras.applications import VGG16
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.callbacks import ReduceLROnPlateau
from sklearn.utils.class_weight import compute_class_weight
from tensorflow.keras.callbacks import EarlyStopping
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
```

##Veri Setini Yükleme

[13]:
```
# Veri yolu
train_dir = "/content/Yilan/train"
test_dir = "/content/Yilan/test"
```

##Model Parametreleri ve Eğitim Ayarları

[ ]:
```
# Model parametreleri
input_shape = (224, 224, 3)  # Görüntü boyutları
num_classes = 1  # Binary sınıflandırma (Venomous/Non-Venomous)
batch_size = 32
```

#Veri Artırma ve Veri Yükleme (Data Augmentation & Generators)

[ ]:
```
# Eğitim için veri artırma (Data Augmentation)
train_datagen = ImageDataGenerator(
    rescale=1.0/255,                # Normalizasyon [0, 1]
    rotation_range=20,             # Görüntü döndürme (derece cinsinden)
    width_shift_range=0.2,         # Görüntü genişliğini kaydırma
    height_shift_range=0.2,        # Görüntü yüksekliğini kaydırma
    shear_range=0.2,               # Kesme işlemi
    zoom_range=0.2,                # Yakınlaştırma
```

```
    horizontal_flip=True,          # Yatay çevirme
    fill_mode='nearest'            # Doldurma modu (boşlukları doldur)
)

test_datagen = ImageDataGenerator(rescale=1.0/255)  # Sadece normalize

train_generator = train_datagen.flow_from_directory(
    train_dir,
    target_size=(224, 224),
    batch_size=batch_size,
    class_mode='binary',  # Binary sınıflandırma için
    classes=['Non_Venomous', 'Venomous']  # Klasörleri manuel olarak belirtin
)

test_generator = test_datagen.flow_from_directory(
    test_dir,
    target_size=(224, 224),
    batch_size=batch_size,
    class_mode='binary',  # Binary sınıflandırma için
    classes=['Non_Venomous', 'Venomous']  # Klasörleri manuel olarak belirtin
)
```

##Eğitim ve Test Verisi Kontrolleri

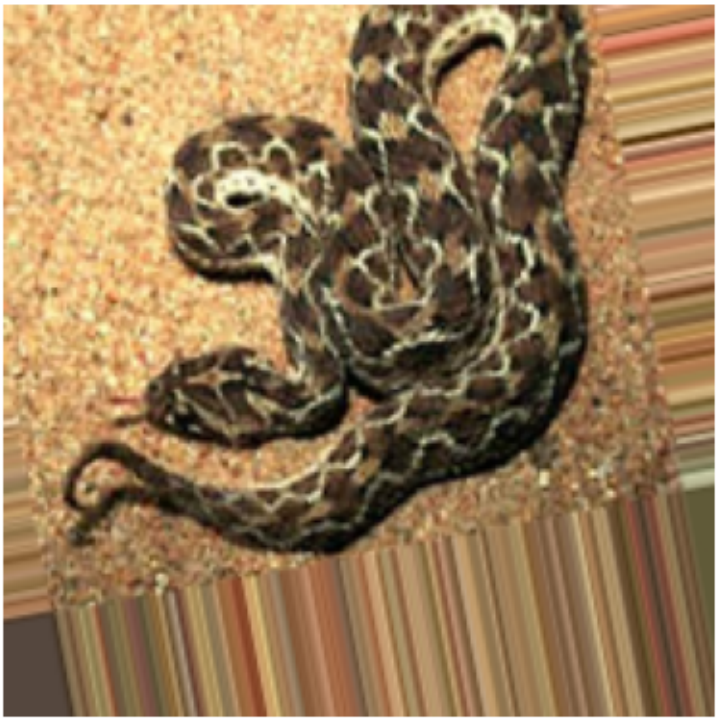
[ ]:
```
# Eğitim etiketlerini kontrol et
print("Eğitim Etiketleri:")
print(np.unique(train_generator.classes))

# Doğrulama etiketlerini kontrol et
print("Doğrulama Etiketleri:")
print(np.unique(test_generator.classes))

# Eğitim setindeki sınıf dağılımı
unique, counts = np.unique(train_generator.classes, return_counts=True)
print("Eğitim Sınıf Dağılımı:")
print(dict(zip(unique, counts)))

# Doğrulama setindeki sınıf dağılımı
unique, counts = np.unique(test_generator.classes, return_counts=True)
print("Doğrulama Sınıf Dağılımı:")
print(dict(zip(unique, counts)))

print("Sınıf İsimleri (Eğitim):")
print(train_generator.class_indices)
```

```
Eğitim Etiketleri:
[0 1]
Doğrulama Etiketleri:
```

```
[0 1]
Eğitim Sınıf Dağılımı:
{0: 715, 1: 1060}
Doğrulama Sınıf Dağılımı:
{0: 128, 1: 141}
Sınıf İsimleri (Eğitim):
{'Non_Venomous': 0, 'Venomous': 1}
```

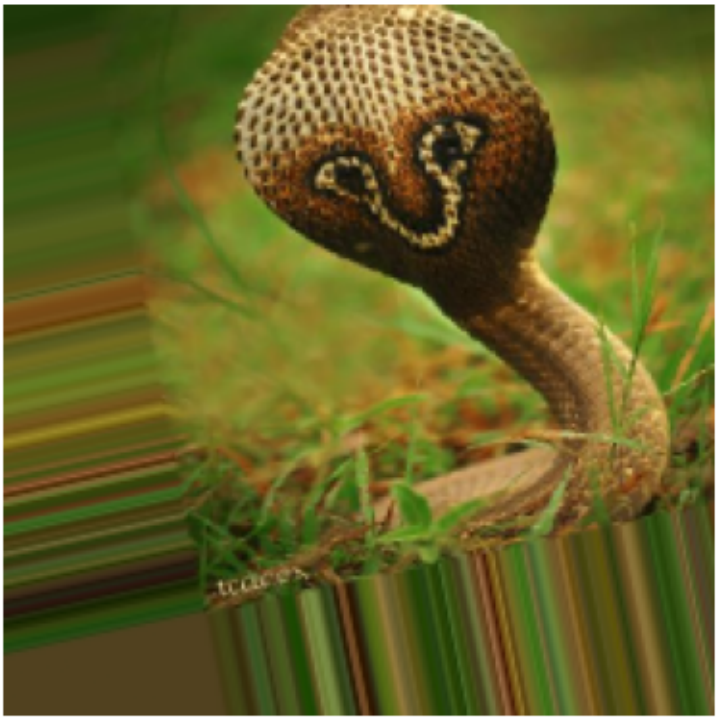
Etiket: 0.0

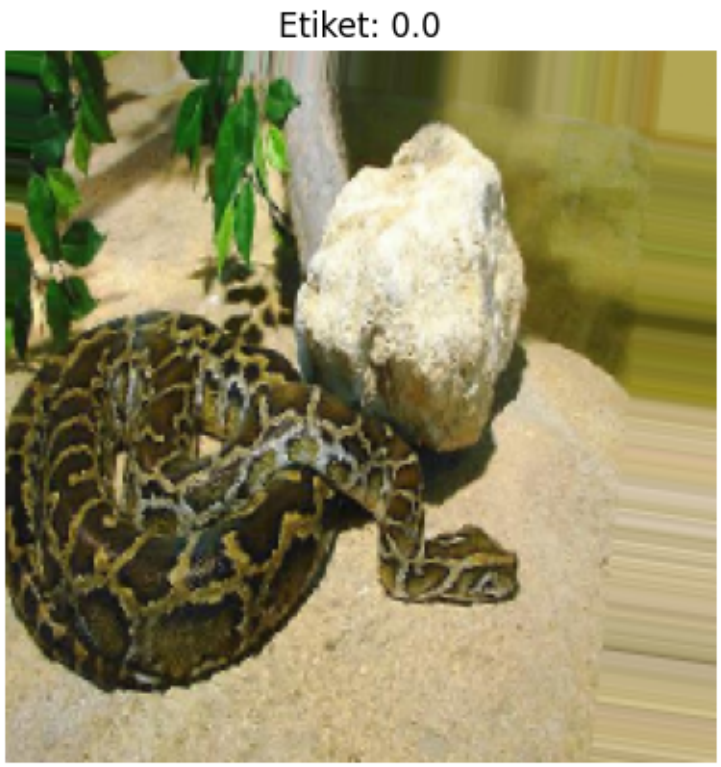
Etiket: 1.0



Etiket: 0.0

Etiket: 1.0





##Veri Görselleştirme - Eğitim Verisi Örnekleri

[ ]:
```
# Eğitim setinden birkaç örnek al
x_batch, y_batch = next(train_generator)

# İlk 5 görüntüyü ve etiketlerini göster
for i in range(5):
    plt.imshow(x_batch[i])
    plt.title(f"Etiket: {y_batch[i]}")
    plt.axis('off')
    plt.show()
```

##Model Oluşturma - ResNet50 Transfer Learning

[ ]:
```
from tensorflow.keras.applications import ResNet50
# ResNet50 Modeli Oluşturma
def create_resnet50_model(input_shape, num_classes):
    base_model = ResNet50(weights='imagenet', include_top=False,␣
 ↪input_shape=input_shape)

    # Taban katmanları dondur
    for layer in base_model.layers:
        layer.trainable = False
```

```
    # Yeni katmanlar ekle
    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(128, activation='relu')(x)
    x = Dropout(0.5)(x)
    predictions = Dense(1, activation='sigmoid')(x)

    model = Model(inputs=base_model.input, outputs=predictions)
    return model

# Modeli oluştur
resnet_model = create_resnet50_model(input_shape=input_shape,␣
 ↪num_classes=num_classes)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5
94765736/94765736             0s
0us/step
Epoch 1/50
```

```
/usr/local/lib/python3.10/dist-
packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:122:
UserWarning: Your `PyDataset` class should call `super().__init__(**kwargs)` in
its constructor. `**kwargs` can include `workers`, `use_multiprocessing`,
`max_queue_size`. Do not pass these arguments to `fit()`, as they will be
ignored.
  self._warn_if_super_not_called()
```

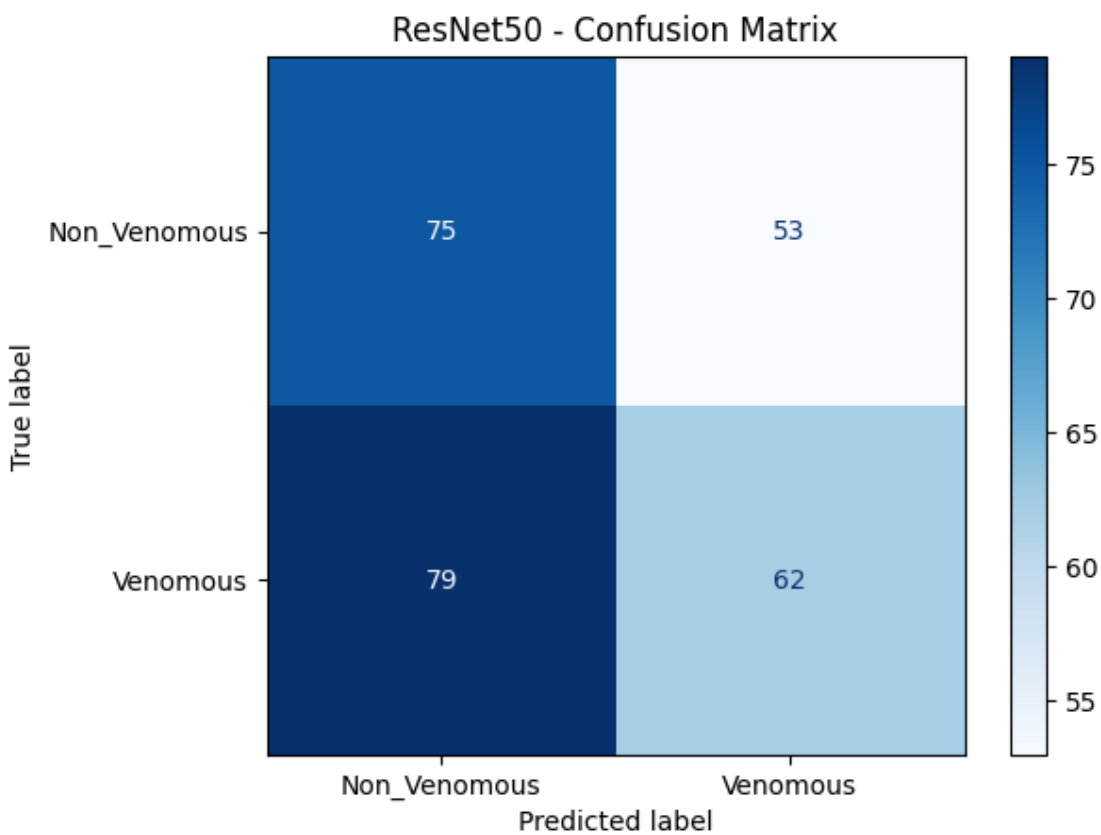
```
56/56             70s 852ms/step -
accuracy: 0.5132 - loss: 0.7967 - val_accuracy: 0.4647 - val_loss: 0.6962 -
learning_rate: 1.0000e-04
Epoch 2/50
56/56             32s 517ms/step -
accuracy: 0.5302 - loss: 0.7333 - val_accuracy: 0.5130 - val_loss: 0.6935 -
learning_rate: 1.0000e-04
Epoch 3/50
56/56             32s 505ms/step -
accuracy: 0.5103 - loss: 0.7110 - val_accuracy: 0.5093 - val_loss: 0.6961 -
learning_rate: 1.0000e-04
Epoch 4/50
56/56             30s 480ms/step -
accuracy: 0.4683 - loss: 0.7101 - val_accuracy: 0.4907 - val_loss: 0.6911 -
learning_rate: 1.0000e-04
Epoch 5/50
56/56             31s 499ms/step -
accuracy: 0.4896 - loss: 0.7002 - val_accuracy: 0.5167 - val_loss: 0.6875 -
learning_rate: 1.0000e-04
```

```
Epoch 6/50
56/56              41s 504ms/step -
accuracy: 0.5077 - loss: 0.6982 - val_accuracy: 0.5242 - val_loss: 0.6874 -
learning_rate: 1.0000e-04
Epoch 7/50
56/56              31s 495ms/step -
accuracy: 0.5303 - loss: 0.6928 - val_accuracy: 0.5242 - val_loss: 0.6860 -
learning_rate: 1.0000e-04
Epoch 8/50
56/56              39s 481ms/step -
accuracy: 0.5731 - loss: 0.6850 - val_accuracy: 0.5428 - val_loss: 0.6887 -
learning_rate: 1.0000e-04
Epoch 9/50
56/56              44s 513ms/step -
accuracy: 0.4892 - loss: 0.6879 - val_accuracy: 0.5725 - val_loss: 0.6850 -
learning_rate: 1.0000e-04
Epoch 10/50
56/56              32s 512ms/step -
accuracy: 0.5125 - loss: 0.6874 - val_accuracy: 0.5279 - val_loss: 0.6862 -
learning_rate: 1.0000e-04
Epoch 11/50
56/56              39s 475ms/step -
accuracy: 0.5317 - loss: 0.6839 - val_accuracy: 0.5204 - val_loss: 0.6862 -
learning_rate: 1.0000e-04
Epoch 12/50
56/56              29s 447ms/step -
accuracy: 0.5196 - loss: 0.6897 - val_accuracy: 0.5279 - val_loss: 0.6857 -
learning_rate: 1.0000e-04
Epoch 13/50
56/56              36s 370ms/step -
accuracy: 0.4948 - loss: 0.6916 - val_accuracy: 0.5465 - val_loss: 0.6849 -
learning_rate: 5.0000e-05
Epoch 14/50
56/56              24s 372ms/step -
accuracy: 0.5511 - loss: 0.6821 - val_accuracy: 0.5390 - val_loss: 0.6846 -
learning_rate: 5.0000e-05
Epoch 15/50
56/56              40s 382ms/step -
accuracy: 0.5232 - loss: 0.6861 - val_accuracy: 0.5576 - val_loss: 0.6824 -
learning_rate: 5.0000e-05
Epoch 16/50
56/56              23s 367ms/step -
accuracy: 0.5168 - loss: 0.6914 - val_accuracy: 0.5465 - val_loss: 0.6834 -
learning_rate: 5.0000e-05
Epoch 17/50
56/56              42s 391ms/step -
accuracy: 0.5456 - loss: 0.6884 - val_accuracy: 0.5428 - val_loss: 0.6831 -
learning_rate: 5.0000e-05
```

```
Epoch 18/50
56/56              41s 390ms/step -
accuracy: 0.5090 - loss: 0.6942 - val_accuracy: 0.5428 - val_loss: 0.6824 -
learning_rate: 5.0000e-05
Epoch 19/50
56/56              25s 396ms/step -
accuracy: 0.5554 - loss: 0.6845 - val_accuracy: 0.5465 - val_loss: 0.6820 -
learning_rate: 2.5000e-05
Epoch 20/50
56/56              41s 397ms/step -
accuracy: 0.5342 - loss: 0.6858 - val_accuracy: 0.5651 - val_loss: 0.6818 -
learning_rate: 2.5000e-05
Epoch 21/50
56/56              24s 391ms/step -
accuracy: 0.5177 - loss: 0.6814 - val_accuracy: 0.5576 - val_loss: 0.6820 -
learning_rate: 2.5000e-05
Epoch 22/50
56/56              24s 386ms/step -
accuracy: 0.5067 - loss: 0.6857 - val_accuracy: 0.5576 - val_loss: 0.6820 -
learning_rate: 2.5000e-05
Epoch 23/50
56/56              41s 389ms/step -
accuracy: 0.5324 - loss: 0.6877 - val_accuracy: 0.5688 - val_loss: 0.6815 -
learning_rate: 2.5000e-05
Epoch 24/50
56/56              24s 390ms/step -
accuracy: 0.5165 - loss: 0.6837 - val_accuracy: 0.5651 - val_loss: 0.6815 -
learning_rate: 2.5000e-05
Epoch 25/50
56/56              24s 378ms/step -
accuracy: 0.5234 - loss: 0.6933 - val_accuracy: 0.5613 - val_loss: 0.6809 -
learning_rate: 2.5000e-05
Epoch 26/50
56/56              42s 390ms/step -
accuracy: 0.4863 - loss: 0.6944 - val_accuracy: 0.5465 - val_loss: 0.6812 -
learning_rate: 2.5000e-05
Epoch 27/50
56/56              41s 383ms/step -
accuracy: 0.4939 - loss: 0.6906 - val_accuracy: 0.5651 - val_loss: 0.6807 -
learning_rate: 2.5000e-05
Epoch 28/50
56/56              40s 366ms/step -
accuracy: 0.5219 - loss: 0.6829 - val_accuracy: 0.5688 - val_loss: 0.6808 -
learning_rate: 2.5000e-05
Epoch 29/50
56/56              27s 376ms/step -
accuracy: 0.5525 - loss: 0.6838 - val_accuracy: 0.5502 - val_loss: 0.6807 -
learning_rate: 2.5000e-05
```

```
Epoch 30/50
56/56              37s 364ms/step -
accuracy: 0.5209 - loss: 0.6843 - val_accuracy: 0.5465 - val_loss: 0.6807 -
learning_rate: 2.5000e-05
Epoch 31/50
56/56              24s 368ms/step -
accuracy: 0.5392 - loss: 0.6813 - val_accuracy: 0.5613 - val_loss: 0.6805 -
learning_rate: 1.2500e-05
Epoch 32/50
56/56              40s 383ms/step -
accuracy: 0.4844 - loss: 0.6899 - val_accuracy: 0.5539 - val_loss: 0.6803 -
learning_rate: 1.2500e-05
Epoch 33/50
56/56              25s 384ms/step -
accuracy: 0.5046 - loss: 0.6911 - val_accuracy: 0.5576 - val_loss: 0.6803 -
learning_rate: 1.2500e-05
Epoch 34/50
56/56              24s 369ms/step -
accuracy: 0.5134 - loss: 0.6877 - val_accuracy: 0.5613 - val_loss: 0.6804 -
learning_rate: 1.2500e-05
Epoch 35/50
56/56              40s 375ms/step -
accuracy: 0.5257 - loss: 0.6841 - val_accuracy: 0.5502 - val_loss: 0.6803 -
learning_rate: 1.2500e-05
Epoch 36/50
56/56              24s 371ms/step -
accuracy: 0.5504 - loss: 0.6835 - val_accuracy: 0.5465 - val_loss: 0.6804 -
learning_rate: 6.2500e-06
Epoch 37/50
56/56              24s 379ms/step -
accuracy: 0.5366 - loss: 0.6832 - val_accuracy: 0.5576 - val_loss: 0.6803 -
learning_rate: 6.2500e-06
Epoch 38/50
56/56              25s 395ms/step -
accuracy: 0.5200 - loss: 0.6917 - val_accuracy: 0.5725 - val_loss: 0.6802 -
learning_rate: 6.2500e-06
Epoch 39/50
56/56              41s 379ms/step -
accuracy: 0.5131 - loss: 0.6926 - val_accuracy: 0.5613 - val_loss: 0.6801 -
learning_rate: 3.1250e-06
Epoch 40/50
56/56              40s 365ms/step -
accuracy: 0.5198 - loss: 0.6778 - val_accuracy: 0.5613 - val_loss: 0.6802 -
learning_rate: 3.1250e-06
Epoch 41/50
56/56              24s 367ms/step -
accuracy: 0.5300 - loss: 0.6810 - val_accuracy: 0.5613 - val_loss: 0.6802 -
learning_rate: 3.1250e-06
```

```
Epoch 42/50
56/56              25s 395ms/step -
accuracy: 0.5143 - loss: 0.6869 - val_accuracy: 0.5576 - val_loss: 0.6803 -
learning_rate: 3.1250e-06
Epoch 43/50
56/56              40s 365ms/step -
accuracy: 0.5007 - loss: 0.6888 - val_accuracy: 0.5651 - val_loss: 0.6802 -
learning_rate: 1.5625e-06
Epoch 44/50
56/56              24s 371ms/step -
accuracy: 0.5140 - loss: 0.6891 - val_accuracy: 0.5688 - val_loss: 0.6802 -
learning_rate: 1.5625e-06

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.

9/9              1s 79ms/step -
accuracy: 0.5541 - loss: 0.6749
Test Loss: 0.6801440119743347, Test Accuracy: 0.5613383054733276
9/9              8s 433ms/step
```



###Model Derleme ve Eğitim - ResNet50

[ ]:
```
# Modeli derle
resnet_model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Callbacks tanımlama
lr_scheduler_res = ReduceLROnPlateau(
    monitor='val_loss',
    factor=0.5,
    patience=3,
    min_lr=1e-7
)
early_stop_res = EarlyStopping(
    monitor='val_loss',
    patience=5,
    restore_best_weights=True
)

# Sınıf ağırlıkları
class_weights_array = compute_class_weight(
    class_weight='balanced',
    classes=np.unique(train_generator.classes),
    y=train_generator.classes
)
class_weights = {i: weight for i, weight in enumerate(class_weights_array)}

# Modeli eğitme
history_res = resnet_model.fit(
    train_generator,
    epochs=50,
    validation_data=test_generator,
    class_weight=class_weights,
    callbacks=[lr_scheduler_res, early_stop_res]
)
```

###Model Kaydetme ve Değerlendirme - ResNet50

[ ]:
```
# Modeli kaydet
resnet_model.save("resnet50_snake_model.h5")

# Modeli değerlendirme
```

```
test_loss_res, test_accuracy_res = resnet_model.evaluate(test_generator)
print(f"Test Loss: {test_loss_res}, Test Accuracy: {test_accuracy_res}")
```

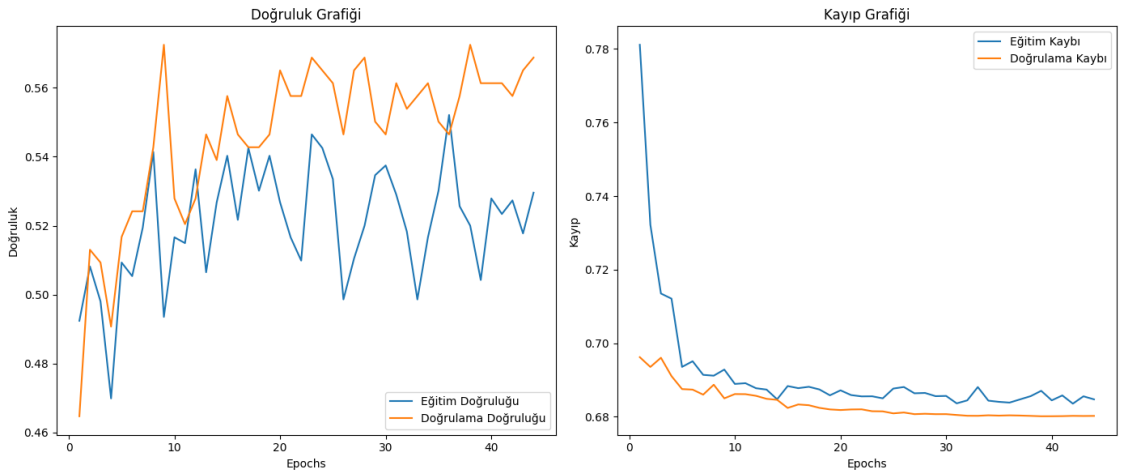
###Confusion Matrix - ResNet50

[ ]:
```
# Confusion Matrix için tahminler
test_labels = test_generator.classes  # Test verisinin gerçek etiketleri
class_names = list(test_generator.class_indices.keys())  # Sınıf isimleri

resnet_predictions = resnet_model.predict(test_generator)
resnet_predicted_labels = np.round(resnet_predictions).astype(int)  # Tahminler␣
↪binary olduğu için yuvarlama

# Confusion Matrix Çizimi
cm = confusion_matrix(test_labels, resnet_predicted_labels)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=class_names)
disp.plot(cmap=plt.cm.Blues)
plt.title("ResNet50 - Confusion Matrix")
plt.show()
```

###Eğitim Süreci Grafikleri - ResNet50

[ ]:
```
train_acc_res = history_res.history['accuracy']
val_acc_res = history_res.history['val_accuracy']
train_loss_res = history_res.history['loss']
val_loss_res = history_res.history['val_loss']

epochs_res = range(1, len(train_acc_res) + 1)

plt.figure(figsize=(14, 6))

# Doğruluk grafiği
plt.subplot(1, 2, 1)
plt.plot(epochs_res, train_acc_res, label='Eğitim Doğruluğu')
plt.plot(epochs_res, val_acc_res, label='Doğrulama Doğruluğu')
plt.title('Doğruluk Grafiği')
plt.xlabel('Epochs')
plt.ylabel('Doğruluk')
plt.legend()

# Kayıp grafiği
plt.subplot(1, 2, 2)
plt.plot(epochs_res, train_loss_res, label='Eğitim Kaybı')
plt.plot(epochs_res, val_loss_res, label='Doğrulama Kaybı')
plt.title('Kayıp Grafiği')
plt.xlabel('Epochs')
plt.ylabel('Kayıp')
```

```
plt.legend()

plt.tight_layout()
plt.show()
```



##Transfer Learning - VGG16

[ ]:
```
# Transfer Learning Modeli
def create_transfer_learning_model(input_shape, num_classes):
    base_model = VGG16(weights='imagenet', include_top=False,␣
 ↪input_shape=input_shape)

    # Yeni katmanlar ekleyin
    x = base_model.output
    x = GlobalAveragePooling2D()(x)
    x = Dense(128, activation='relu')(x)
    x = Dropout(0.4)(x)
    predictions = Dense(num_classes, activation='sigmoid' if num_classes == 1␣
 ↪else 'softmax')(x)

    model = Model(inputs=base_model.input, outputs=predictions)

    # Önceden eğitilmiş katmanları dondur
    for layer in base_model.layers:
        layer.trainable = False

    return model
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-
applications/vgg16/vgg16_weights_tf_dim_ordering_tf_kernels_notop.h5
58889256/58889256              0s
```

```
0us/step
```

###Model Derleme ve Eğitim - VGG16

[ ]:
```
# Modeli oluştur
cnn_model = create_transfer_learning_model(input_shape, num_classes)

# İlk birkaç epoch'tan sonra dondurmayı kaldırın
for layer in cnn_model.layers[:15]:  # İlk 15 katmanı eğitime aç
    layer.trainable = True

cnn_model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-5),  # Daha düşük␣
 ↪öğrenme hızı
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Model özeti
cnn_model.summary()

# Model eğitimi
history_vgg = cnn_model.fit(
    train_generator,
    epochs=50,  # Fine-tuning için birkaç epoch (ör. 10-20) ekleyebilirsiniz
    validation_data=test_generator,
    class_weight=class_weights,  # Sınıf ağırlıkları
    callbacks=[lr_scheduler_vgg,early_stop_vgg]  # Öğrenme oranını dinamik␣
 ↪olarak ayarlamak için
)
```

[ ]:
```

```

[ ]:
```
# Sınıf ağırlıklarını hesaplama
class_weights_array = compute_class_weight(
    class_weight='balanced',
    classes=np.unique(train_generator.classes),
    y=train_generator.classes
)

# Sınıf ağırlıklarını dict formatına dönüştürme
class_weights = {i: weight for i, weight in enumerate(class_weights_array)}

# ReduceLROnPlateau tanımlama
lr_scheduler_vgg = ReduceLROnPlateau(
    monitor='val_loss',   # Hangi metriği takip edecek
    factor=0.5,           # Öğrenme hızını yarıya indir
    patience=3,           # Kaç epoch bekleyecek
```

```
    min_lr=1e-6            # Minimum öğrenme hızı
)

early_stop_vgg = EarlyStopping(
    monitor='val_loss',
    patience=2,  # Bekleme süresini 10 epoch'a çıkardık
    min_delta=0.001,
    restore_best_weights=True
)
```

```
Epoch 1/50
56/56              118s 1s/step -
accuracy: 0.5261 - loss: 0.7184 - val_accuracy: 0.6283 - val_loss: 0.6704 -
learning_rate: 1.0000e-05
Epoch 2/50
56/56              35s 575ms/step -
accuracy: 0.5465 - loss: 0.7046 - val_accuracy: 0.6691 - val_loss: 0.6479 -
learning_rate: 1.0000e-05
Epoch 3/50
56/56              35s 570ms/step -
accuracy: 0.5676 - loss: 0.6740 - val_accuracy: 0.6580 - val_loss: 0.6193 -
learning_rate: 1.0000e-05
Epoch 4/50
56/56              35s 569ms/step -
accuracy: 0.6269 - loss: 0.6362 - val_accuracy: 0.7175 - val_loss: 0.5833 -
learning_rate: 1.0000e-05
Epoch 5/50
56/56              34s 565ms/step -
accuracy: 0.6726 - loss: 0.6064 - val_accuracy: 0.6357 - val_loss: 0.6307 -
learning_rate: 1.0000e-05
Epoch 6/50
56/56              36s 574ms/step -
accuracy: 0.6796 - loss: 0.5945 - val_accuracy: 0.7398 - val_loss: 0.5439 -
learning_rate: 1.0000e-05
Epoch 7/50
56/56              41s 570ms/step -
accuracy: 0.7046 - loss: 0.5557 - val_accuracy: 0.7249 - val_loss: 0.5262 -
learning_rate: 1.0000e-05
Epoch 8/50
56/56              34s 569ms/step -
accuracy: 0.7259 - loss: 0.5476 - val_accuracy: 0.7398 - val_loss: 0.5187 -
learning_rate: 1.0000e-05
Epoch 9/50
56/56              35s 570ms/step -
accuracy: 0.7631 - loss: 0.5028 - val_accuracy: 0.7361 - val_loss: 0.5120 -
learning_rate: 1.0000e-05
Epoch 10/50
```

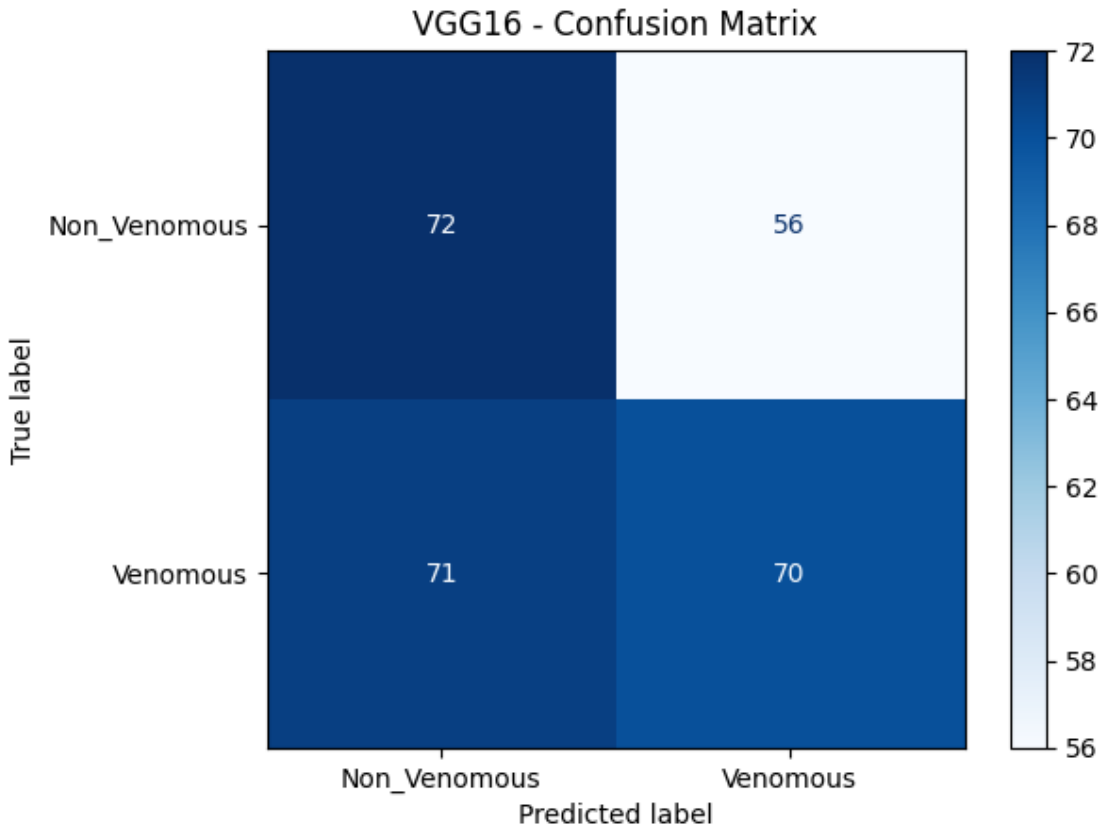
```
56/56              36s 573ms/step -
accuracy: 0.7491 - loss: 0.5171 - val_accuracy: 0.7844 - val_loss: 0.4607 -
learning_rate: 1.0000e-05
Epoch 11/50
56/56              41s 573ms/step -
accuracy: 0.7707 - loss: 0.4730 - val_accuracy: 0.7770 - val_loss: 0.4736 -
learning_rate: 1.0000e-05
Epoch 12/50
56/56              40s 573ms/step -
accuracy: 0.7966 - loss: 0.4539 - val_accuracy: 0.7844 - val_loss: 0.4453 -
learning_rate: 1.0000e-05
Epoch 13/50
56/56              34s 569ms/step -
accuracy: 0.7714 - loss: 0.4624 - val_accuracy: 0.7695 - val_loss: 0.4575 -
learning_rate: 1.0000e-05
Epoch 14/50
56/56              42s 577ms/step -
accuracy: 0.8071 - loss: 0.4389 - val_accuracy: 0.7955 - val_loss: 0.4271 -
learning_rate: 1.0000e-05
Epoch 15/50
56/56              41s 568ms/step -
accuracy: 0.7883 - loss: 0.4464 - val_accuracy: 0.7807 - val_loss: 0.4735 -
learning_rate: 1.0000e-05
Epoch 16/50
56/56              41s 574ms/step -
accuracy: 0.8131 - loss: 0.4116 - val_accuracy: 0.7658 - val_loss: 0.4756 -
learning_rate: 1.0000e-05
```

0.0.1 Model Değerlendirme ve Confusion Matrix - VGG16

[ ]:
```
# Test verileri ile değerlendirme
test_loss_vgg, test_accuracy_vgg = cnn_model.evaluate(test_generator)
print(f"Test Loss: {test_loss_vgg}, Test Accuracy: {test_accuracy_vgg}")

# Confusion Matrix için tahminler
test_labels = test_generator.classes  # Test verisinin gerçek etiketleri
class_names = list(test_generator.class_indices.keys())  # Sınıf isimleri

vgg16_predictions = cnn_model.predict(test_generator)
vgg16_predicted_labels = np.round(vgg16_predictions).astype(int)  # Binary␣
↪sınıflandırma için yuvarlama

# Confusion Matrix Çizimi
cm_vgg = confusion_matrix(test_labels, vgg16_predicted_labels)
disp_vgg = ConfusionMatrixDisplay(confusion_matrix=cm_vgg,␣
↪display_labels=class_names)
disp_vgg.plot(cmap=plt.cm.Blues)
```

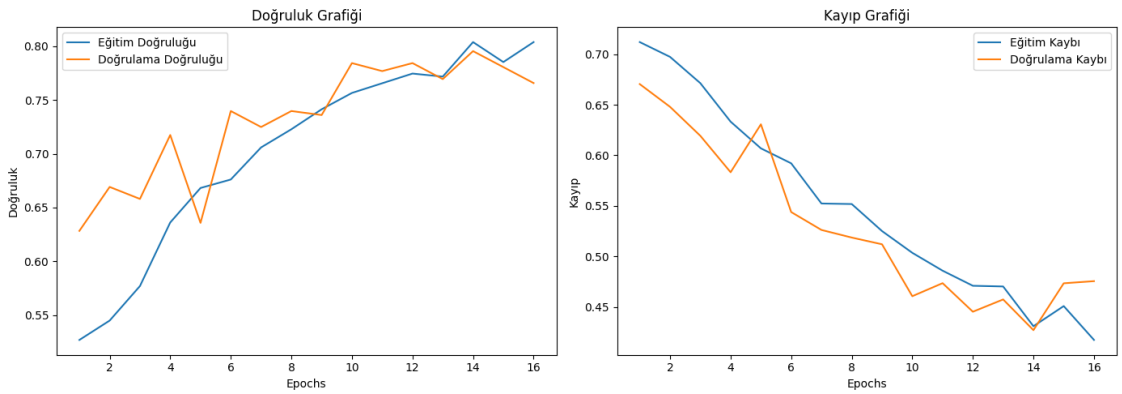
```
plt.title("VGG16 - Confusion Matrix")
plt.show()
```

```
9/9              2s 169ms/step
```



[ ]:
```
import matplotlib.pyplot as plt

# Eğitim ve doğrulama metriklerini alma
train_accuracy_vgg = history_vgg.history['accuracy']
val_accuracy_vgg = history_vgg.history['val_accuracy']
train_loss_vgg  = history_vgg.history['loss']
val_loss_vgg = history_vgg.history['val_loss']

# Epoch sayısını belirleme
epochs_vgg = range(1, len(train_accuracy_vgg) + 1)

# 1. Doğruluk grafiği
plt.figure(figsize=(14, 5))

plt.subplot(1, 2, 1)
```

```
plt.plot(epochs_vgg, train_accuracy_vgg, label='Eğitim Doğruluğu')
plt.plot(epochs_vgg, val_accuracy_vgg, label='Doğrulama Doğruluğu')
plt.title('Doğruluk Grafiği')
plt.xlabel('Epochs')
plt.ylabel('Doğruluk')
plt.legend()

# 2. Kayıp grafiği
plt.subplot(1, 2, 2)
plt.plot(epochs_vgg, train_loss_vgg, label='Eğitim Kaybı')
plt.plot(epochs_vgg, val_loss_vgg, label='Doğrulama Kaybı')
plt.title('Kayıp Grafiği')
plt.xlabel('Epochs')
plt.ylabel('Kayıp')
plt.legend()

plt.tight_layout()
plt.show()
```



1 MobilNetV2 modeli

[ ]:
```
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.models import Model
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Dropout
import tensorflow as tf

# MobilNetV2 modeli
def create_mobilenetv2_model(input_shape, num_classes):
    base_model = MobileNetV2(weights='imagenet', include_top=False,␣
 ↪input_shape=input_shape)

    # Taban katmanlarını dondur
```

```
    for layer in base_model.layers:
        layer.trainable = False

    # Yeni katmanlar ekleyin
    x = base_model.output
    x = GlobalAveragePooling2D()(x)  # Havuzlama katmanı
    x = Dense(64, activation='relu', kernel_regularizer=tf.keras.regularizers.
 ↪l2(0.01))(x)
    x = Dropout(0.4)(x)
    predictions = Dense(1, activation='sigmoid')(x)  # Binary sınıflandırma

    model = Model(inputs=base_model.input, outputs=predictions)
    return model

# Modeli oluştur
mobilenet_model = create_mobilenetv2_model(input_shape=(224, 224, 3),␣
 ↪num_classes=1)

# Modeli derle
mobilenet_model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applicatio
ns/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_224_no_top.h
5
9406464/9406464            0s
0us/step
```

[ ]:
```
# Model özeti
mobilenet_model.summary()
```

```
Model: "functional_2"


 Layer (type)                    Output Shape                  Param #   Connected␣
 ↪to

 input_layer_2                   (None, 224, 224, 3)                0   -         ␣
 ↪
 (InputLayer)                                                             ␣
 ↪

 Conv1 (Conv2D)                  (None, 112, 112, 32)             864   ␣
 ↪input_layer_2[0][0]
```

```
bn_Conv1                    (None, 112, 112, 32)                128 ␣
↪Conv1[0][0]
(BatchNormalization)                                                ␣
↪

Conv1_relu (ReLU)           (None, 112, 112, 32)                  0 ␣
↪bn_Conv1[0][0]

expanded_conv_depthwise     (None, 112, 112, 32)                288 ␣
↪Conv1_relu[0][0]
(DepthwiseConv2D)                                                   ␣
↪

expanded_conv_depthwise_…   (None, 112, 112, 32)                128 ␣
↪expanded_conv_depthwi…
(BatchNormalization)                                                ␣
↪

expanded_conv_depthwise_…   (None, 112, 112, 32)                  0 ␣
↪expanded_conv_depthwi…
(ReLU)                                                              ␣
↪

expanded_conv_project       (None, 112, 112, 16)                512 ␣
↪expanded_conv_depthwi…
(Conv2D)                                                            ␣
↪

expanded_conv_project_BN    (None, 112, 112, 16)                 64 ␣
↪expanded_conv_project…
(BatchNormalization)                                                ␣
↪

block_1_expand (Conv2D)     (None, 112, 112, 96)              1,536 ␣
↪expanded_conv_project…

block_1_expand_BN           (None, 112, 112, 96)                384 ␣
↪block_1_expand[0][0]
(BatchNormalization)                                                ␣
↪

block_1_expand_relu         (None, 112, 112, 96)                  0 ␣
↪block_1_expand_BN[0][…
(ReLU)                                                              ␣
↪
```

```
block_1_pad                  (None, 113, 113, 96)                 0 ␣
↪block_1_expand_relu[0…
(ZeroPadding2D)                                                     ␣
↪

block_1_depthwise            (None, 56, 56, 96)                 864 ␣
↪block_1_pad[0][0]
(DepthwiseConv2D)                                                   ␣
↪

block_1_depthwise_BN         (None, 56, 56, 96)                 384 ␣
↪block_1_depthwise[0][…
(BatchNormalization)                                                ␣
↪

block_1_depthwise_relu       (None, 56, 56, 96)                   0 ␣
↪block_1_depthwise_BN[…
(ReLU)                                                              ␣
↪

block_1_project (Conv2D)     (None, 56, 56, 24)               2,304 ␣
↪block_1_depthwise_rel…

block_1_project_BN           (None, 56, 56, 24)                  96 ␣
↪block_1_project[0][0]
(BatchNormalization)                                                ␣
↪

block_2_expand (Conv2D)      (None, 56, 56, 144)              3,456 ␣
↪block_1_project_BN[0]…

block_2_expand_BN            (None, 56, 56, 144)                576 ␣
↪block_2_expand[0][0]
(BatchNormalization)                                                ␣
↪

block_2_expand_relu          (None, 56, 56, 144)                  0 ␣
↪block_2_expand_BN[0][…
(ReLU)                                                              ␣
↪

block_2_depthwise            (None, 56, 56, 144)              1,296 ␣
↪block_2_expand_relu[0…
(DepthwiseConv2D)                                                   ␣
↪
```

```
block_2_depthwise_BN        (None, 56, 56, 144)                576 ␣
↪block_2_depthwise[0][…
(BatchNormalization)                                                ␣
↪

block_2_depthwise_relu      (None, 56, 56, 144)                  0 ␣
↪block_2_depthwise_BN[…
(ReLU)                                                              ␣
↪

block_2_project (Conv2D)    (None, 56, 56, 24)                3,456 ␣
↪block_2_depthwise_rel…

block_2_project_BN          (None, 56, 56, 24)                   96 ␣
↪block_2_project[0][0]
(BatchNormalization)                                                ␣
↪

block_2_add (Add)           (None, 56, 56, 24)                    0 ␣
↪block_1_project_BN[0]…
                                                                    ␣
↪block_2_project_BN[0]…

block_3_expand (Conv2D)     (None, 56, 56, 144)               3,456 ␣
↪block_2_add[0][0]

block_3_expand_BN           (None, 56, 56, 144)                  576 ␣
↪block_3_expand[0][0]
(BatchNormalization)                                                ␣
↪

block_3_expand_relu         (None, 56, 56, 144)                    0 ␣
↪block_3_expand_BN[0][…
(ReLU)                                                              ␣
↪

block_3_pad                 (None, 57, 57, 144)                    0 ␣
↪block_3_expand_relu[0…
(ZeroPadding2D)                                                     ␣
↪

block_3_depthwise           (None, 28, 28, 144)               1,296 ␣
↪block_3_pad[0][0]
(DepthwiseConv2D)                                                   ␣
↪
```

```
block_3_depthwise_BN        (None, 28, 28, 144)                576 ␣
↪block_3_depthwise[0][…
(BatchNormalization)                                                ␣
↪

block_3_depthwise_relu      (None, 28, 28, 144)                  0 ␣
↪block_3_depthwise_BN[…
(ReLU)                                                              ␣
↪

block_3_project (Conv2D)    (None, 28, 28, 32)               4,608 ␣
↪block_3_depthwise_rel…

block_3_project_BN          (None, 28, 28, 32)                 128 ␣
↪block_3_project[0][0]
(BatchNormalization)                                                ␣
↪

block_4_expand (Conv2D)     (None, 28, 28, 192)              6,144 ␣
↪block_3_project_BN[0]…

block_4_expand_BN           (None, 28, 28, 192)                768 ␣
↪block_4_expand[0][0]
(BatchNormalization)                                                ␣
↪

block_4_expand_relu         (None, 28, 28, 192)                  0 ␣
↪block_4_expand_BN[0][…
(ReLU)                                                              ␣
↪

block_4_depthwise           (None, 28, 28, 192)              1,728 ␣
↪block_4_expand_relu[0…
(DepthwiseConv2D)                                                   ␣
↪

block_4_depthwise_BN        (None, 28, 28, 192)                768 ␣
↪block_4_depthwise[0][…
(BatchNormalization)                                                ␣
↪

block_4_depthwise_relu      (None, 28, 28, 192)                  0 ␣
↪block_4_depthwise_BN[…
(ReLU)                                                              ␣
↪
```

```
block_4_project (Conv2D)     (None, 28, 28, 32)          6,144 ␣
↪block_4_depthwise_rel…

block_4_project_BN           (None, 28, 28, 32)            128 ␣
↪block_4_project[0][0]
(BatchNormalization)                                             ␣
↪

block_4_add (Add)            (None, 28, 28, 32)              0 ␣
↪block_3_project_BN[0]…
                                                                 ␣
↪block_4_project_BN[0]…

block_5_expand (Conv2D)      (None, 28, 28, 192)         6,144 ␣
↪block_4_add[0][0]

block_5_expand_BN            (None, 28, 28, 192)           768 ␣
↪block_5_expand[0][0]
(BatchNormalization)                                             ␣
↪

block_5_expand_relu          (None, 28, 28, 192)             0 ␣
↪block_5_expand_BN[0][…
(ReLU)                                                           ␣
↪

block_5_depthwise            (None, 28, 28, 192)         1,728 ␣
↪block_5_expand_relu[0…
(DepthwiseConv2D)                                                ␣
↪

block_5_depthwise_BN         (None, 28, 28, 192)           768 ␣
↪block_5_depthwise[0][…
(BatchNormalization)                                             ␣
↪

block_5_depthwise_relu       (None, 28, 28, 192)             0 ␣
↪block_5_depthwise_BN[…
(ReLU)                                                           ␣
↪

block_5_project (Conv2D)     (None, 28, 28, 32)          6,144 ␣
↪block_5_depthwise_rel…
```

```
block_5_project_BN          (None, 28, 28, 32)                128 ␣
↪block_5_project[0][0]
(BatchNormalization)                                                ␣
↪

block_5_add (Add)           (None, 28, 28, 32)                  0 ␣
↪block_4_add[0][0],
                                                                  ␣
↪block_5_project_BN[0]…

block_6_expand (Conv2D)     (None, 28, 28, 192)             6,144 ␣
↪block_5_add[0][0]

block_6_expand_BN           (None, 28, 28, 192)               768 ␣
↪block_6_expand[0][0]
(BatchNormalization)                                                ␣
↪

block_6_expand_relu         (None, 28, 28, 192)                 0 ␣
↪block_6_expand_BN[0][…
(ReLU)                                                              ␣
↪

block_6_pad                 (None, 29, 29, 192)                 0 ␣
↪block_6_expand_relu[0…
(ZeroPadding2D)                                                     ␣
↪

block_6_depthwise           (None, 14, 14, 192)             1,728 ␣
↪block_6_pad[0][0]
(DepthwiseConv2D)                                                   ␣
↪

block_6_depthwise_BN        (None, 14, 14, 192)               768 ␣
↪block_6_depthwise[0][…
(BatchNormalization)                                                ␣
↪

block_6_depthwise_relu      (None, 14, 14, 192)                 0 ␣
↪block_6_depthwise_BN[…
(ReLU)                                                              ␣
↪

block_6_project (Conv2D)    (None, 14, 14, 64)             12,288 ␣
↪block_6_depthwise_rel…
```

```
block_6_project_BN          (None, 14, 14, 64)                256 ␣
↪block_6_project[0][0]
(BatchNormalization)                                                ␣
↪

block_7_expand (Conv2D)     (None, 14, 14, 384)            24,576 ␣
↪block_6_project_BN[0]…

block_7_expand_BN           (None, 14, 14, 384)             1,536 ␣
↪block_7_expand[0][0]
(BatchNormalization)                                                ␣
↪

block_7_expand_relu         (None, 14, 14, 384)                 0 ␣
↪block_7_expand_BN[0][…
(ReLU)                                                              ␣
↪

block_7_depthwise           (None, 14, 14, 384)             3,456 ␣
↪block_7_expand_relu[0…
(DepthwiseConv2D)                                                   ␣
↪

block_7_depthwise_BN        (None, 14, 14, 384)             1,536 ␣
↪block_7_depthwise[0][…
(BatchNormalization)                                                ␣
↪

block_7_depthwise_relu      (None, 14, 14, 384)                 0 ␣
↪block_7_depthwise_BN[…
(ReLU)                                                              ␣
↪

block_7_project (Conv2D)    (None, 14, 14, 64)             24,576 ␣
↪block_7_depthwise_rel…

block_7_project_BN          (None, 14, 14, 64)                256 ␣
↪block_7_project[0][0]
(BatchNormalization)                                                ␣
↪

block_7_add (Add)           (None, 14, 14, 64)                  0 ␣
↪block_6_project_BN[0]…
                                                                    ␣
↪block_7_project_BN[0]…
```

```
block_8_expand (Conv2D)      (None, 14, 14, 384)          24,576 ␣
↪block_7_add[0][0]

block_8_expand_BN            (None, 14, 14, 384)           1,536 ␣
↪block_8_expand[0][0]
(BatchNormalization)                                                ␣
↪

block_8_expand_relu          (None, 14, 14, 384)               0 ␣
↪block_8_expand_BN[0][…
(ReLU)                                                              ␣
↪

block_8_depthwise            (None, 14, 14, 384)           3,456 ␣
↪block_8_expand_relu[0…
(DepthwiseConv2D)                                                   ␣
↪

block_8_depthwise_BN         (None, 14, 14, 384)           1,536 ␣
↪block_8_depthwise[0][…
(BatchNormalization)                                                ␣
↪

block_8_depthwise_relu       (None, 14, 14, 384)               0 ␣
↪block_8_depthwise_BN[…
(ReLU)                                                              ␣
↪

block_8_project (Conv2D)     (None, 14, 14, 64)           24,576 ␣
↪block_8_depthwise_rel…

block_8_project_BN           (None, 14, 14, 64)              256 ␣
↪block_8_project[0][0]
(BatchNormalization)                                                ␣
↪

block_8_add (Add)            (None, 14, 14, 64)                0 ␣
↪block_7_add[0][0],
                                                                    ␣
↪block_8_project_BN[0]…

block_9_expand (Conv2D)      (None, 14, 14, 384)          24,576 ␣
↪block_8_add[0][0]

block_9_expand_BN            (None, 14, 14, 384)           1,536 ␣
↪block_9_expand[0][0]
```

```
(BatchNormalization)                                                    ␣
↪

block_9_expand_relu          (None, 14, 14, 384)                   0  ␣
↪block_9_expand_BN[0][…
(ReLU)                                                                  ␣
↪

block_9_depthwise            (None, 14, 14, 384)               3,456  ␣
↪block_9_expand_relu[0…
(DepthwiseConv2D)                                                       ␣
↪

block_9_depthwise_BN         (None, 14, 14, 384)               1,536  ␣
↪block_9_depthwise[0][…
(BatchNormalization)                                                    ␣
↪

block_9_depthwise_relu       (None, 14, 14, 384)                   0  ␣
↪block_9_depthwise_BN[…
(ReLU)                                                                  ␣
↪

block_9_project (Conv2D)     (None, 14, 14, 64)               24,576  ␣
↪block_9_depthwise_rel…

block_9_project_BN           (None, 14, 14, 64)                  256  ␣
↪block_9_project[0][0]
(BatchNormalization)                                                    ␣
↪

block_9_add (Add)            (None, 14, 14, 64)                    0  ␣
↪block_8_add[0][0],
                                                                        ␣
↪block_9_project_BN[0]…

block_10_expand (Conv2D)     (None, 14, 14, 384)              24,576  ␣
↪block_9_add[0][0]

block_10_expand_BN           (None, 14, 14, 384)               1,536  ␣
↪block_10_expand[0][0]
(BatchNormalization)                                                    ␣
↪

block_10_expand_relu         (None, 14, 14, 384)                   0  ␣
↪block_10_expand_BN[0]…
```

```
(ReLU)                                                                  ␣
↪

block_10_depthwise          (None, 14, 14, 384)              3,456  ␣
↪block_10_expand_relu[…
(DepthwiseConv2D)                                                       ␣
↪

block_10_depthwise_BN       (None, 14, 14, 384)              1,536  ␣
↪block_10_depthwise[0]…
(BatchNormalization)                                                    ␣
↪

block_10_depthwise_relu     (None, 14, 14, 384)                  0  ␣
↪block_10_depthwise_BN…
(ReLU)                                                                  ␣
↪

block_10_project (Conv2D)   (None, 14, 14, 96)              36,864  ␣
↪block_10_depthwise_re…

block_10_project_BN         (None, 14, 14, 96)                 384  ␣
↪block_10_project[0][0]
(BatchNormalization)                                                    ␣
↪

block_11_expand (Conv2D)    (None, 14, 14, 576)             55,296  ␣
↪block_10_project_BN[0…

block_11_expand_BN          (None, 14, 14, 576)              2,304  ␣
↪block_11_expand[0][0]
(BatchNormalization)                                                    ␣
↪

block_11_expand_relu        (None, 14, 14, 576)                  0  ␣
↪block_11_expand_BN[0]…
(ReLU)                                                                  ␣
↪

block_11_depthwise          (None, 14, 14, 576)              5,184  ␣
↪block_11_expand_relu[…
(DepthwiseConv2D)                                                       ␣
↪

block_11_depthwise_BN       (None, 14, 14, 576)              2,304  ␣
↪block_11_depthwise[0]…
```

```
(BatchNormalization)                                                    ␣
↪

block_11_depthwise_relu     (None, 14, 14, 576)                  0  ␣
↪block_11_depthwise_BN…
(ReLU)                                                                  ␣
↪

block_11_project (Conv2D)   (None, 14, 14, 96)               55,296  ␣
↪block_11_depthwise_re…

block_11_project_BN         (None, 14, 14, 96)                  384  ␣
↪block_11_project[0][0]
(BatchNormalization)                                                    ␣
↪

block_11_add (Add)          (None, 14, 14, 96)                    0  ␣
↪block_10_project_BN[0…
                                                                        ␣
↪block_11_project_BN[0…

block_12_expand (Conv2D)    (None, 14, 14, 576)              55,296  ␣
↪block_11_add[0][0]

block_12_expand_BN          (None, 14, 14, 576)               2,304  ␣
↪block_12_expand[0][0]
(BatchNormalization)                                                    ␣
↪

block_12_expand_relu        (None, 14, 14, 576)                   0  ␣
↪block_12_expand_BN[0]…
(ReLU)                                                                  ␣
↪

block_12_depthwise          (None, 14, 14, 576)               5,184  ␣
↪block_12_expand_relu[…
(DepthwiseConv2D)                                                       ␣
↪

block_12_depthwise_BN       (None, 14, 14, 576)               2,304  ␣
↪block_12_depthwise[0]…
(BatchNormalization)                                                    ␣
↪

block_12_depthwise_relu     (None, 14, 14, 576)                   0  ␣
↪block_12_depthwise_BN…
```

```
(ReLU)                                                                  ␣
↪

block_12_project (Conv2D)   (None, 14, 14, 96)              55,296  ␣
↪block_12_depthwise_re…

block_12_project_BN         (None, 14, 14, 96)                 384  ␣
↪block_12_project[0][0]
(BatchNormalization)                                                    ␣
↪

block_12_add (Add)          (None, 14, 14, 96)                   0  ␣
↪block_11_add[0][0],
                                                                        ␣
↪block_12_project_BN[0…

block_13_expand (Conv2D)    (None, 14, 14, 576)             55,296  ␣
↪block_12_add[0][0]

block_13_expand_BN          (None, 14, 14, 576)              2,304  ␣
↪block_13_expand[0][0]
(BatchNormalization)                                                    ␣
↪

block_13_expand_relu        (None, 14, 14, 576)                  0  ␣
↪block_13_expand_BN[0]…
(ReLU)                                                                  ␣
↪

block_13_pad                (None, 15, 15, 576)                  0  ␣
↪block_13_expand_relu[…
(ZeroPadding2D)                                                         ␣
↪

block_13_depthwise          (None, 7, 7, 576)                5,184  ␣
↪block_13_pad[0][0]
(DepthwiseConv2D)                                                       ␣
↪

block_13_depthwise_BN       (None, 7, 7, 576)                2,304  ␣
↪block_13_depthwise[0]…
(BatchNormalization)                                                    ␣
↪

block_13_depthwise_relu     (None, 7, 7, 576)                    0  ␣
↪block_13_depthwise_BN…
```

```
(ReLU)                                                                  ␣
↪

block_13_project (Conv2D)   (None, 7, 7, 160)              92,160  ␣
↪block_13_depthwise_re…

block_13_project_BN         (None, 7, 7, 160)                 640  ␣
↪block_13_project[0][0]
(BatchNormalization)                                                    ␣
↪

block_14_expand (Conv2D)    (None, 7, 7, 960)             153,600  ␣
↪block_13_project_BN[0…

block_14_expand_BN          (None, 7, 7, 960)               3,840  ␣
↪block_14_expand[0][0]
(BatchNormalization)                                                    ␣
↪

block_14_expand_relu        (None, 7, 7, 960)                   0  ␣
↪block_14_expand_BN[0]…
(ReLU)                                                                  ␣
↪

block_14_depthwise          (None, 7, 7, 960)               8,640  ␣
↪block_14_expand_relu[…
(DepthwiseConv2D)                                                       ␣
↪

block_14_depthwise_BN       (None, 7, 7, 960)               3,840  ␣
↪block_14_depthwise[0]…
(BatchNormalization)                                                    ␣
↪

block_14_depthwise_relu     (None, 7, 7, 960)                   0  ␣
↪block_14_depthwise_BN…
(ReLU)                                                                  ␣
↪

block_14_project (Conv2D)   (None, 7, 7, 160)             153,600  ␣
↪block_14_depthwise_re…

block_14_project_BN         (None, 7, 7, 160)                 640  ␣
↪block_14_project[0][0]
(BatchNormalization)                                                    ␣
↪
```

```
block_14_add (Add)          (None, 7, 7, 160)                    0 ␣
↪block_13_project_BN[0…
                                                                  ␣
↪block_14_project_BN[0…

block_15_expand (Conv2D)    (None, 7, 7, 960)              153,600 ␣
↪block_14_add[0][0]

block_15_expand_BN          (None, 7, 7, 960)                3,840 ␣
↪block_15_expand[0][0]
(BatchNormalization)                                                ␣
↪

block_15_expand_relu        (None, 7, 7, 960)                    0 ␣
↪block_15_expand_BN[0]…
(ReLU)                                                              ␣
↪

block_15_depthwise          (None, 7, 7, 960)                8,640 ␣
↪block_15_expand_relu[…
(DepthwiseConv2D)                                                   ␣
↪

block_15_depthwise_BN       (None, 7, 7, 960)                3,840 ␣
↪block_15_depthwise[0]…
(BatchNormalization)                                                ␣
↪

block_15_depthwise_relu     (None, 7, 7, 960)                    0 ␣
↪block_15_depthwise_BN…
(ReLU)                                                              ␣
↪

block_15_project (Conv2D)   (None, 7, 7, 160)              153,600 ␣
↪block_15_depthwise_re…

block_15_project_BN         (None, 7, 7, 160)                  640 ␣
↪block_15_project[0][0]
(BatchNormalization)                                                ␣
↪

block_15_add (Add)          (None, 7, 7, 160)                    0 ␣
↪block_14_add[0][0],
                                                                  ␣
↪block_15_project_BN[0…
```

```
block_16_expand (Conv2D)     (None, 7, 7, 960)          153,600 ␣
↪block_15_add[0][0]

block_16_expand_BN           (None, 7, 7, 960)            3,840 ␣
↪block_16_expand[0][0]
(BatchNormalization)                                                ␣
↪

block_16_expand_relu         (None, 7, 7, 960)                0 ␣
↪block_16_expand_BN[0]…
(ReLU)                                                              ␣
↪

block_16_depthwise           (None, 7, 7, 960)            8,640 ␣
↪block_16_expand_relu[…
(DepthwiseConv2D)                                                   ␣
↪

block_16_depthwise_BN        (None, 7, 7, 960)            3,840 ␣
↪block_16_depthwise[0]…
(BatchNormalization)                                                ␣
↪

block_16_depthwise_relu      (None, 7, 7, 960)                0 ␣
↪block_16_depthwise_BN…
(ReLU)                                                              ␣
↪

block_16_project (Conv2D)    (None, 7, 7, 320)          307,200 ␣
↪block_16_depthwise_re…

block_16_project_BN          (None, 7, 7, 320)            1,280 ␣
↪block_16_project[0][0]
(BatchNormalization)                                                ␣
↪

Conv_1 (Conv2D)              (None, 7, 7, 1280)         409,600 ␣
↪block_16_project_BN[0…

Conv_1_bn                    (None, 7, 7, 1280)           5,120 ␣
↪Conv_1[0][0]
(BatchNormalization)                                                ␣
↪
```

```
 out_relu (ReLU)             (None, 7, 7, 1280)                  0 ␣
↪Conv_1_bn[0][0]

 global_average_pooling2d…   (None, 1280)                        0 ␣
↪out_relu[0][0]
 (GlobalAveragePooling2D)                                          ␣
↪

 dense_4 (Dense)             (None, 64)                     81,984 ␣
↪global_average_poolin…

 dropout_2 (Dropout)         (None, 64)                          0 ␣
↪dense_4[0][0]

 dense_5 (Dense)             (None, 1)                          65 ␣
↪dropout_2[0][0]


Total params: 2,340,033 (8.93 MB)


Trainable params: 82,049 (320.50 KB)


Non-trainable params: 2,257,984 (8.61 MB)
```

[ ]:
```
# Model özeti
mobilenet_model.summary()

# Sınıf ağırlıklarını hesaplama
class_weights_array = compute_class_weight(
    class_weight='balanced',
    classes=np.unique(train_generator.classes),
    y=train_generator.classes
)
class_weights = {i: weight for i, weight in enumerate(class_weights_array)}

# Callbacks tanımlama
lr_scheduler_mobile = ReduceLROnPlateau(
    monitor='val_loss',    # Hangi metriği takip edecek
    factor=0.5,            # Öğrenme hızını yarıya indir
    patience=3,            # Kaç epoch bekleyecek
    min_lr=1e-6            # Minimum öğrenme hızı
)

early_stop_mobile = EarlyStopping(
```

```
    monitor='val_loss',
    patience=7,  # Bekleme süresini 10 epoch'a çıkardık
    min_delta=0.001,
    restore_best_weights=True
)




# Model eğitimi
history_mobile = mobilenet_model.fit(
    train_generator,
    epochs=50,
    validation_data=test_generator,
    class_weight=class_weights,
    callbacks=[lr_scheduler_mobile, early_stop_mobile]  # ReduceLROnPlateau ve␣
 ↪EarlyStopping
)
```

```
Epoch 1/50
56/56              43s 533ms/step -
accuracy: 0.5661 - loss: 1.9807 - val_accuracy: 0.6952 - val_loss: 1.7510 -
learning_rate: 1.0000e-04
Epoch 2/50
56/56              26s 376ms/step -
accuracy: 0.6560 - loss: 1.7671 - val_accuracy: 0.7361 - val_loss: 1.6333 -
learning_rate: 1.0000e-04
Epoch 3/50
56/56              22s 351ms/step -
accuracy: 0.6804 - loss: 1.6757 - val_accuracy: 0.7584 - val_loss: 1.5473 -
learning_rate: 1.0000e-04
Epoch 4/50
56/56              42s 371ms/step -
accuracy: 0.7343 - loss: 1.5669 - val_accuracy: 0.8030 - val_loss: 1.4706 -
learning_rate: 1.0000e-04
Epoch 5/50
56/56              23s 370ms/step -
accuracy: 0.7349 - loss: 1.5141 - val_accuracy: 0.8141 - val_loss: 1.4057 -
learning_rate: 1.0000e-04
Epoch 6/50
56/56              22s 345ms/step -
accuracy: 0.7625 - loss: 1.4364 - val_accuracy: 0.8141 - val_loss: 1.3547 -
learning_rate: 1.0000e-04
Epoch 7/50
56/56              23s 343ms/step -
accuracy: 0.7716 - loss: 1.3771 - val_accuracy: 0.8327 - val_loss: 1.2947 -
learning_rate: 1.0000e-04
```

```
Epoch 8/50
56/56              23s 367ms/step -
accuracy: 0.7794 - loss: 1.3309 - val_accuracy: 0.8216 - val_loss: 1.2480 -
learning_rate: 1.0000e-04
Epoch 9/50
56/56              22s 351ms/step -
accuracy: 0.7756 - loss: 1.2743 - val_accuracy: 0.8327 - val_loss: 1.2041 -
learning_rate: 1.0000e-04
Epoch 10/50
56/56              42s 369ms/step -
accuracy: 0.7900 - loss: 1.2225 - val_accuracy: 0.8401 - val_loss: 1.1667 -
learning_rate: 1.0000e-04
Epoch 11/50
56/56              41s 347ms/step -
accuracy: 0.8028 - loss: 1.1952 - val_accuracy: 0.8290 - val_loss: 1.1302 -
learning_rate: 1.0000e-04
Epoch 12/50
56/56              40s 348ms/step -
accuracy: 0.8091 - loss: 1.1316 - val_accuracy: 0.8216 - val_loss: 1.0994 -
learning_rate: 1.0000e-04
Epoch 13/50
56/56              23s 354ms/step -
accuracy: 0.8139 - loss: 1.1268 - val_accuracy: 0.8401 - val_loss: 1.0633 -
learning_rate: 1.0000e-04
Epoch 14/50
56/56              40s 348ms/step -
accuracy: 0.8309 - loss: 1.0691 - val_accuracy: 0.8290 - val_loss: 1.0389 -
learning_rate: 1.0000e-04
Epoch 15/50
56/56              23s 350ms/step -
accuracy: 0.8143 - loss: 1.0538 - val_accuracy: 0.8401 - val_loss: 1.0049 -
learning_rate: 1.0000e-04
Epoch 16/50
56/56              40s 347ms/step -
accuracy: 0.8029 - loss: 1.0341 - val_accuracy: 0.8439 - val_loss: 0.9845 -
learning_rate: 1.0000e-04
Epoch 17/50
56/56              23s 363ms/step -
accuracy: 0.8258 - loss: 0.9955 - val_accuracy: 0.8364 - val_loss: 0.9584 -
learning_rate: 1.0000e-04
Epoch 18/50
56/56              23s 374ms/step -
accuracy: 0.7929 - loss: 1.0012 - val_accuracy: 0.8364 - val_loss: 0.9422 -
learning_rate: 1.0000e-04
Epoch 19/50
56/56              22s 350ms/step -
accuracy: 0.8197 - loss: 0.9500 - val_accuracy: 0.8253 - val_loss: 0.9187 -
learning_rate: 1.0000e-04
```

```
Epoch 20/50
56/56              42s 376ms/step -
accuracy: 0.8320 - loss: 0.9192 - val_accuracy: 0.8401 - val_loss: 0.9006 -
learning_rate: 1.0000e-04
Epoch 21/50
56/56              24s 381ms/step -
accuracy: 0.8389 - loss: 0.8999 - val_accuracy: 0.8401 - val_loss: 0.8744 -
learning_rate: 1.0000e-04
Epoch 22/50
56/56              22s 353ms/step -
accuracy: 0.8315 - loss: 0.8850 - val_accuracy: 0.8364 - val_loss: 0.8655 -
learning_rate: 1.0000e-04
Epoch 23/50
56/56              24s 355ms/step -
accuracy: 0.8312 - loss: 0.8698 - val_accuracy: 0.8364 - val_loss: 0.8458 -
learning_rate: 1.0000e-04
Epoch 24/50
56/56              23s 371ms/step -
accuracy: 0.8507 - loss: 0.8266 - val_accuracy: 0.8439 - val_loss: 0.8248 -
learning_rate: 1.0000e-04
Epoch 25/50
56/56              41s 351ms/step -
accuracy: 0.8575 - loss: 0.8117 - val_accuracy: 0.8476 - val_loss: 0.8080 -
learning_rate: 1.0000e-04
Epoch 26/50
56/56              23s 377ms/step -
accuracy: 0.8557 - loss: 0.7947 - val_accuracy: 0.8476 - val_loss: 0.7941 -
learning_rate: 1.0000e-04
Epoch 27/50
56/56              23s 361ms/step -
accuracy: 0.8554 - loss: 0.7797 - val_accuracy: 0.8327 - val_loss: 0.7810 -
learning_rate: 1.0000e-04
Epoch 28/50
56/56              42s 374ms/step -
accuracy: 0.8462 - loss: 0.7820 - val_accuracy: 0.8439 - val_loss: 0.7680 -
learning_rate: 1.0000e-04
Epoch 29/50
56/56              41s 348ms/step -
accuracy: 0.8557 - loss: 0.7524 - val_accuracy: 0.8401 - val_loss: 0.7564 -
learning_rate: 1.0000e-04
Epoch 30/50
56/56              23s 373ms/step -
accuracy: 0.8592 - loss: 0.7413 - val_accuracy: 0.8476 - val_loss: 0.7441 -
learning_rate: 1.0000e-04
Epoch 31/50
56/56              41s 358ms/step -
accuracy: 0.8496 - loss: 0.7297 - val_accuracy: 0.8439 - val_loss: 0.7336 -
learning_rate: 1.0000e-04
```

```
Epoch 32/50
56/56              23s 375ms/step -
accuracy: 0.8492 - loss: 0.7318 - val_accuracy: 0.8439 - val_loss: 0.7200 -
learning_rate: 1.0000e-04
Epoch 33/50
56/56              22s 357ms/step -
accuracy: 0.8503 - loss: 0.7091 - val_accuracy: 0.8513 - val_loss: 0.7122 -
learning_rate: 1.0000e-04
Epoch 34/50
56/56              44s 421ms/step -
accuracy: 0.8777 - loss: 0.6686 - val_accuracy: 0.8513 - val_loss: 0.7055 -
learning_rate: 1.0000e-04
Epoch 35/50
56/56              24s 384ms/step -
accuracy: 0.8693 - loss: 0.6695 - val_accuracy: 0.8401 - val_loss: 0.6891 -
learning_rate: 1.0000e-04
Epoch 36/50
56/56              23s 365ms/step -
accuracy: 0.8575 - loss: 0.6825 - val_accuracy: 0.8476 - val_loss: 0.6776 -
learning_rate: 1.0000e-04
Epoch 37/50
56/56              23s 352ms/step -
accuracy: 0.8596 - loss: 0.6466 - val_accuracy: 0.8513 - val_loss: 0.6703 -
learning_rate: 1.0000e-04
Epoch 38/50
56/56              41s 362ms/step -
accuracy: 0.8647 - loss: 0.6363 - val_accuracy: 0.8513 - val_loss: 0.6624 -
learning_rate: 1.0000e-04
Epoch 39/50
56/56              23s 354ms/step -
accuracy: 0.8653 - loss: 0.6441 - val_accuracy: 0.8699 - val_loss: 0.6545 -
learning_rate: 1.0000e-04
Epoch 40/50
56/56              41s 365ms/step -
accuracy: 0.8551 - loss: 0.6333 - val_accuracy: 0.8699 - val_loss: 0.6463 -
learning_rate: 1.0000e-04
Epoch 41/50
56/56              41s 373ms/step -
accuracy: 0.8664 - loss: 0.6140 - val_accuracy: 0.8550 - val_loss: 0.6405 -
learning_rate: 1.0000e-04
Epoch 42/50
56/56              22s 360ms/step -
accuracy: 0.8634 - loss: 0.6135 - val_accuracy: 0.8699 - val_loss: 0.6279 -
learning_rate: 1.0000e-04
Epoch 43/50
56/56              22s 351ms/step -
accuracy: 0.8805 - loss: 0.5832 - val_accuracy: 0.8476 - val_loss: 0.6195 -
learning_rate: 1.0000e-04
```

```
Epoch 44/50
56/56              23s 362ms/step -
accuracy: 0.8605 - loss: 0.5966 - val_accuracy: 0.8625 - val_loss: 0.6108 -
learning_rate: 1.0000e-04
Epoch 45/50
56/56              25s 401ms/step -
accuracy: 0.8784 - loss: 0.5827 - val_accuracy: 0.8662 - val_loss: 0.6052 -
learning_rate: 1.0000e-04
Epoch 46/50
56/56              24s 377ms/step -
accuracy: 0.8730 - loss: 0.5704 - val_accuracy: 0.8625 - val_loss: 0.6004 -
learning_rate: 1.0000e-04
Epoch 47/50
56/56              22s 349ms/step -
accuracy: 0.8860 - loss: 0.5692 - val_accuracy: 0.8662 - val_loss: 0.5915 -
learning_rate: 1.0000e-04
Epoch 48/50
56/56              23s 345ms/step -
accuracy: 0.8936 - loss: 0.5528 - val_accuracy: 0.8662 - val_loss: 0.5867 -
learning_rate: 1.0000e-04
Epoch 49/50
56/56              40s 353ms/step -
accuracy: 0.8804 - loss: 0.5378 - val_accuracy: 0.8625 - val_loss: 0.5818 -
learning_rate: 1.0000e-04
Epoch 50/50
56/56              23s 350ms/step -
accuracy: 0.8859 - loss: 0.5360 - val_accuracy: 0.8662 - val_loss: 0.5794 -
learning_rate: 1.0000e-04
```

[ ]:
```python
# Modeli kaydetme
mobilenet_model.save("snake_classification_model.h5")
```

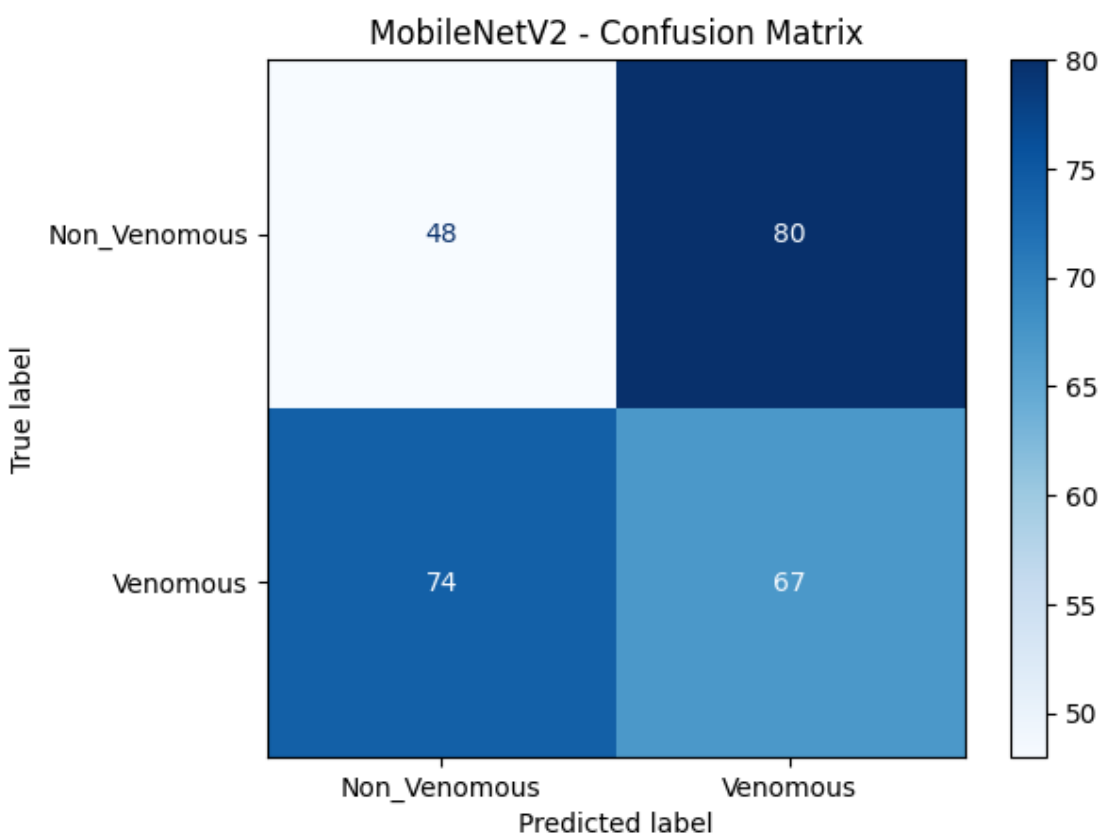
```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.
```

[ ]:
```python
# Test verileri ile değerlendirme
test_loss, test_accuracy = mobilenet_model.evaluate(test_generator)
print(f"Test Loss: {test_loss}, Test Accuracy: {test_accuracy}")
```

```
9/9              1s 64ms/step -
accuracy: 0.8924 - loss: 0.5477
Test Loss: 0.5793561339378357, Test Accuracy: 0.8661710023880005
```

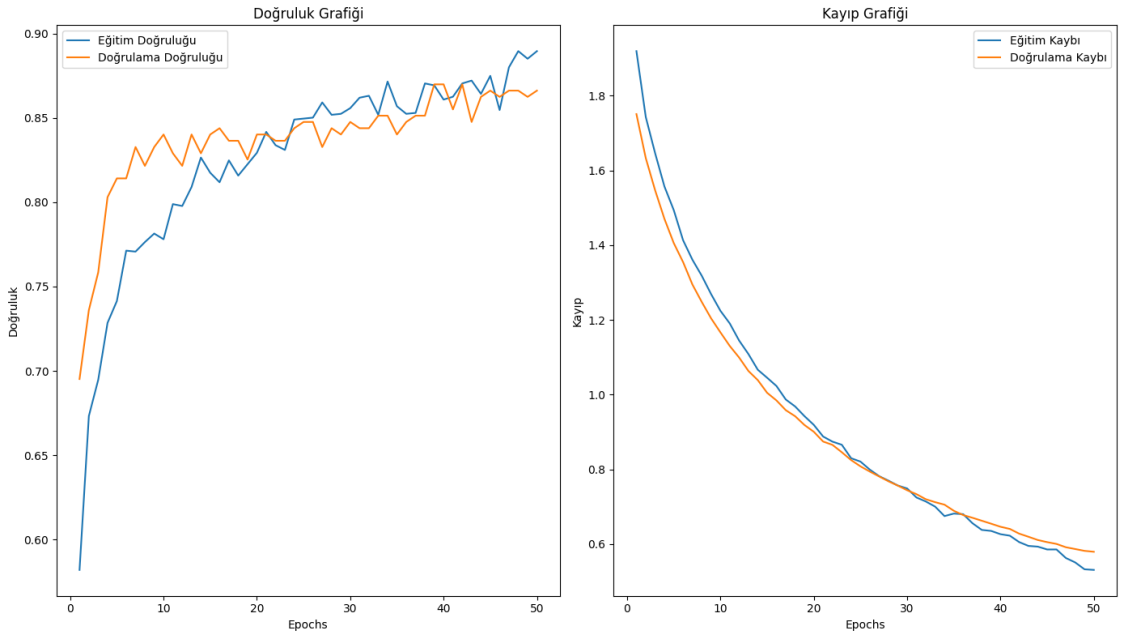
[ ]:
```
# Confusion Matrix için tahminler
test_labels = test_generator.classes  # Test verisinin gerçek etiketleri
class_names = list(test_generator.class_indices.keys())  # Sınıf isimleri

mobilenet_predictions = mobilenet_model.predict(test_generator)
mobilenet_predicted_labels = np.round(mobilenet_predictions).astype(int)  #␣
↪Binary sınıflandırma için yuvarlama

# Confusion Matrix Çizimi
cm_mobile = confusion_matrix(test_labels, mobilenet_predicted_labels)
disp_mobile = ConfusionMatrixDisplay(confusion_matrix=cm_mobile,␣
↪display_labels=class_names)
disp_mobile.plot(cmap=plt.cm.Blues)
plt.title("MobileNetV2 - Confusion Matrix")
plt.show()
```

```
9/9              7s 509ms/step
```



[ ]:
```
import matplotlib.pyplot as plt
```

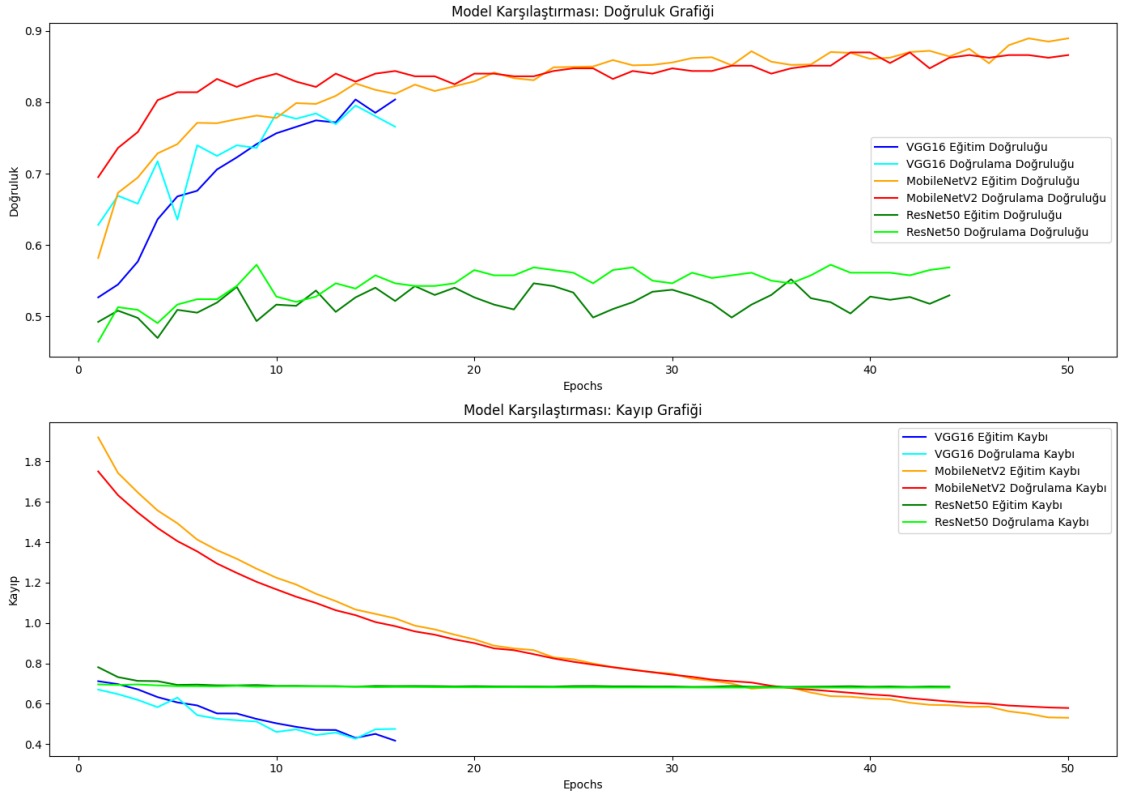
```python
# Eğitim ve doğrulama metriklerini alma
train_acc_mobile = history_mobile.history['accuracy']
val_acc_mobile = history_mobile.history['val_accuracy']
train_loss_mobile = history_mobile.history['loss']
val_loss_mobile = history_mobile.history['val_loss']

# Epoch sayısını belirleme
epochs_mobile = range(1, len(train_acc_mobile) + 1)

# 1. Doğruluk grafiği
plt.figure(figsize=(14, 8))

plt.subplot(1, 2, 1)
plt.plot(epochs_mobile, train_acc_mobile, label='Eğitim Doğruluğu')
plt.plot(epochs_mobile, val_acc_mobile, label='Doğrulama Doğruluğu')
plt.title('Doğruluk Grafiği')
plt.xlabel('Epochs')
plt.ylabel('Doğruluk')
plt.legend()

# 2. Kayıp grafiği
plt.subplot(1, 2, 2)
plt.plot(epochs_mobile, train_loss_mobile, label='Eğitim Kaybı')
plt.plot(epochs_mobile, val_loss_mobile, label='Doğrulama Kaybı')
plt.title('Kayıp Grafiği')
plt.xlabel('Epochs')
plt.ylabel('Kayıp')
plt.legend()

plt.tight_layout()
plt.show()
```



1.1 Model Karşılaştırması (VGG16, MobileNetV2, ResNet50)

[ ]:
```
# Epoch sayısı
epochs_vgg  = range(1, len(train_accuracy_vgg) + 1)
epochs_mobile = range(1, len(train_acc_mobile) + 1)
epochs_res = range(1, len(train_acc_res) + 1)

# Grafik çizimi
plt.figure(figsize=(14, 10))

# 1. Doğruluk Karşılaştırması
plt.subplot(2, 1, 1)
plt.plot(epochs_vgg, train_accuracy_vgg, label='VGG16 Eğitim Doğruluğu',␣
 ↪color='blue')
plt.plot(epochs_vgg, val_accuracy_vgg, label='VGG16 Doğrulama Doğruluğu',␣
 ↪color='cyan')
plt.plot(epochs_mobile, train_acc_mobile, label='MobileNetV2 Eğitim Doğruluğu',␣
 ↪color='orange')
plt.plot(epochs_mobile, val_acc_mobile, label='MobileNetV2 Doğrulama␣
 ↪Doğruluğu', color='red')
plt.plot(epochs_res, train_acc_res, label='ResNet50 Eğitim Doğruluğu',␣
 ↪color='green')
plt.plot(epochs_res, val_acc_res, label='ResNet50 Doğrulama Doğruluğu',␣
 ↪color='lime')
plt.title("Model Karşılaştırması: Doğruluk Grafiği")
```

```
plt.xlabel("Epochs")
plt.ylabel("Doğruluk")
plt.legend()

# 2. Kayıp Karşılaştırması
plt.subplot(2, 1, 2)
plt.plot(epochs_vgg, train_loss_vgg, label='VGG16 Eğitim Kaybı', color='blue')
plt.plot(epochs_vgg, val_loss_vgg, label='VGG16 Doğrulama Kaybı', color='cyan')
plt.plot(epochs_mobile, train_loss_mobile, label='MobileNetV2 Eğitim Kaybı',␣
 ↪color='orange')
plt.plot(epochs_mobile, val_loss_mobile, label='MobileNetV2 Doğrulama Kaybı',␣
 ↪color='red')
plt.plot(epochs_res, train_loss_res, label='ResNet50 Eğitim Kaybı',␣
 ↪color='green')
plt.plot(epochs_res, val_loss_res, label='ResNet50 Doğrulama Kaybı',␣
 ↪color='lime')
plt.title("Model Karşılaştırması: Kayıp Grafiği")
plt.xlabel("Epochs")
plt.ylabel("Kayıp")
plt.legend()

plt.tight_layout()
plt.show()
```



10 Epoch performans karşılaştırılması

30 epoch model karşılaştırması

50 epoch model karşılaştırması #Genel Karşılaştırma Tablosu